

Practical 11

Aim: Use Firebase to perform CRUD operation

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_horizontal"
        android:text="Shree Swami Atmanand Saraswati Institute of Technology"
        android:textSize="20dp"
        android:fontFamily="sans-serif-black"
        android:layout_marginBottom="50dp"/>

    <EditText
        android:id="@+id/name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:hint="Name"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.412"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.081" />

    <EditText
        android:id="@+id/last_name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:hint="Last Name"
        app:layout_constraintBottom_toBottomOf="parent"
```

```

app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.412"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/name"
app:layout_constraintVertical_bias="0.094" />

```

```

<EditText
    android:id="@+id/age"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="number"
    android:hint="Age"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.412"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/last_name"
    app:layout_constraintVertical_bias="0.127" />

```

```

<EditText
    android:id="@+id/phone"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="phone"
    android:hint="Phone number"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.412"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/age"
    app:layout_constraintVertical_bias="0.171" />

```

```

<Button
    android:id="@+id/btn_insert"
    android:layout_width="210dp"
    android:layout_height="54dp"
    android:text="Insert Data"

    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.412"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/phone"
    app:layout_constraintVertical_bias="0.303" />

```

```

<Button
    android:id="@+id/btn_delete"
    android:layout_width="210dp"
    android:layout_height="54dp"
    android:text="Delete Data"

    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.412"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/btn_insert"
    app:layout_constraintVertical_bias="0.181" />

<Button
    android:id="@+id/btn_update"
    android:layout_width="210dp"
    android:layout_height="54dp"
    android:text="Update Data"

    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.412"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/btn_delete"
    app:layout_constraintVertical_bias="0.303" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java:

```

package com.ssasit.pr11;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

```

```

import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;

import java.util.Objects;

public class MainActivity extends AppCompatActivity {
    EditText Name,Last_name,Age,Phone,Email;
    Button InsertData,DeleteData,UpdateData;
    DatabaseReference ref;
    MemberData member;
    Long M_phone;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Name=(EditText) findViewById(R.id.name);
        Last_name=(EditText) findViewById(R.id.last_name);
        Age=(EditText) findViewById(R.id.age);
        Phone=(EditText) findViewById(R.id.phone);
        InsertData=(Button) findViewById(R.id.btn_insert);
        UpdateData=(Button) findViewById(R.id.btn_update);
        DeleteData=(Button) findViewById(R.id.btn_delete);
        ref= FirebaseDatabase.getInstance().getReference().child("Member");
        member = new MemberData();

        InsertData.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                int M_age=Integer.parseInt(Age.getText().toString().trim());
                M_phone=Long.parseLong(Phone.getText().toString().trim());
                member.setName(Name.getText().toString().trim());
                member.setLast_name(Last_name.getText().toString().trim());
                member.setAge(M_age);
                member.setPhone(M_phone);
                if(!Objects.equals(ref.child(M_phone.toString()).getKey(), M_phone.toString())){
                    ref.child(M_phone.toString()).setValue(member);

                    Toast.makeText(MainActivity.this, "Data Inserted",
Toast.LENGTH_SHORT).show();

                }
                else{
                    Toast.makeText(MainActivity.this, "Data Already Available!",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}

```

```

    }
    });

DeleteData.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        M_phone=Long.parseLong(Phone.getText().toString().trim());
        ref.child(M_phone.toString()).removeValue();
        Toast.makeText(MainActivity.this, "Data Deleted",
Toast.LENGTH_SHORT).show();
    }
    });

UpdateData.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        int M_age=Integer.parseInt(Age.getText().toString().trim());
        M_phone=Long.parseLong(Phone.getText().toString().trim());
        member.setName(Name.getText().toString().trim());
        member.setLast_name>Last_name.getText().toString().trim());
        member.setAge(M_age);
        member.setPhone(M_phone);

        ref.child(M_phone.toString()).setValue(member);
        Toast.makeText(MainActivity.this, "Data Updated",
Toast.LENGTH_SHORT).show();
    }
    });

}
}

```

MemberData.java:

```

package com.ssasit.pr11;
public class MemberData {
    private String Name;
    private String Last_name;
    private Integer Age;
    private Long phone;

    public MemberData()
    {

    }
}

```

```
public String getName() {  
    return Name;  
}  
  
public void setName(String name) {  
    Name = name;  
}  
  
public String getLast_name() {  
    return Last_name;  
}  
  
public void setLast_name(String last_name) {  
    Last_name = last_name;  
}  
  
public Integer getAge() {  
    return Age;  
}  
  
public void setAge(Integer age) {  
    Age = age;  
}  
  
public Long getPhone() {  
    return phone;  
}  
  
public void setPhone(Long phone) {  
    this.phone = phone;  
}  
}
```

Output:

 <https://my-application-b65be->  

```
1234567899  
├── age: 21  
├── last_name: "PATEL"  
├── name: "JEET"  
└── phone: 1234567899
```

12:59

Shree Swami Atmanand Saraswati Institute of Technology

Name

Last Name

Age

Phone number

Insert Data

Delete Data

Update Data

12:59

Shree Swami Atmanand Saraswati Institute of Technology

Jeet

Patel

21

1234567899

Insert Data

Delete Data

Update Data

Data Already Available

1:00

Shree Swami Atmanand Saraswati Institute of Technology

Jeet

Patel

21

1234567899

Insert Data

Delete Data

Update Data

Data Deleted

1:00

Shree Swami Atmanand Saraswati Institute of Technology

Jeet

Patel

21

1234567899

Insert Data

Delete Data

Update Data

Data Deleted

Practical 12

Aim: BroadcastReceiver

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tv_status"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.ssasit.pr12;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    TextView status;
    AirplaneModeChangeReceiver airplaneModeChangeReceiver ;

    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    status=findViewById(R.id.tv_status);
    airplaneModeChangeReceiver = new AirplaneModeChangeReceiver(status);
}

@Override
protected void onStart() {
    super.onStart();
    IntentFilter fa = new IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED);
    registerReceiver(airplaneModeChangeReceiver, fa);
}

@Override
protected void onStop() {
    super.onStop();
    unregisterReceiver(airplaneModeChangeReceiver);
}
}

```

AirplaneModeChangeReceiver.java:

```

package com.ssasit.pr12;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.provider.Settings;
import android.widget.TextView;
import android.widget.Toast;

import org.w3c.dom.Text;

public class AirplaneModeChangeReceiver extends BroadcastReceiver {
    TextView tv;
    public AirplaneModeChangeReceiver(TextView s){
        tv=s;
    }
    @Override
    public void onReceive(Context context, Intent intent) {

        if (isAirplaneModeOn(context.getApplicationContext())) {
            Toast.makeText(context, "AirPlane mode is on", Toast.LENGTH_SHORT).show();
            tv.setText("AirPlane mode is on");
        }
    }
}

```

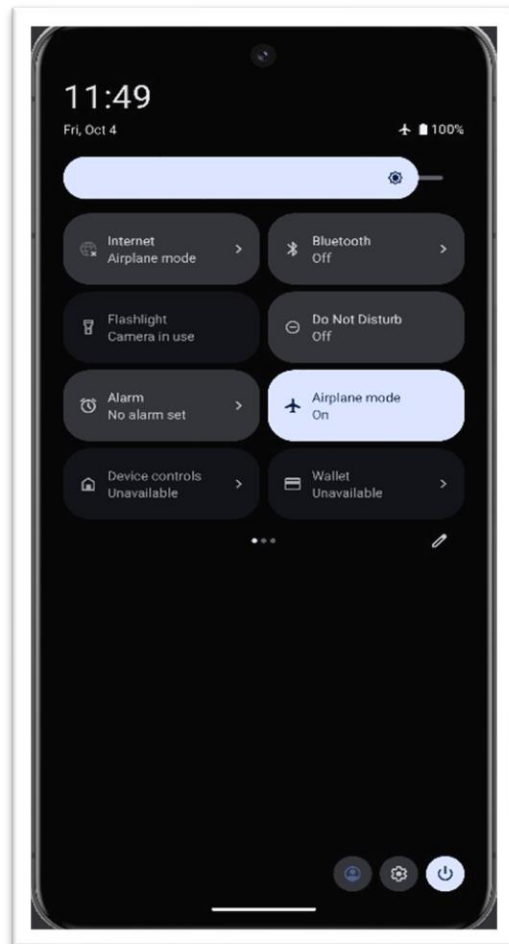
```

    }
    else {
        Toast.makeText(context, "AirPlane mode is off", Toast.LENGTH_SHORT).show();
        tv.setText("AirPlane mode is off");
    }
}

private static boolean isAirplaneModeOn(Context context) {
    return Settings.System.getInt(context.getContentResolver(),
Settings.Global.AIRPLANE_MODE_ON, 0) != 0;
}
}

```

Output:





Practical 13

Aim: Notifications

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"
        android:gravity="center"
        android:text="Notification"
        android:textSize="30sp" />

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_below="@id/textView2"
        android:layout_centerInParent="true"
        android:src="@drawable/notification" />

</RelativeLayout>
```

MainActivity.java:

```
package com.example.pr13;

import android.Manifest;
import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
```

```

import android.content.pm.PackageManager;
import android.graphics.Bitmap;
import android.graphics.drawable.BitmapDrawable;
import android.graphics.drawable.Drawable;
import android.os.Build;
import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.content.ContextCompat;
import androidx.core.content.res.ResourcesCompat;

public class MainActivity extends AppCompatActivity {

    private static final String CHANNEL_ID = "My_Channel_ID";
    private static final int NOTIFICATION_ID = 100;
    private static final int REQUEST_NOTIFICATION_PERMISSION = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Check notification permission for Android 13+ (API level 33)
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (ContextCompat.checkSelfPermission(this,
Manifest.permission.POST_NOTIFICATIONS)
                != PackageManager.PERMISSION_GRANTED) {
                ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.POST_NOTIFICATIONS},
                    REQUEST_NOTIFICATION_PERMISSION);
            } else {
                showNotification();
            }
        } else {
            // For devices below Android 13, show the notification directly
            showNotification();
        }
    }

    private void showNotification() {
        // Get the large icon for the notification
        Drawable drawable = ResourcesCompat.getDrawable(getResources(), R.drawable.img,
null);
        BitmapDrawable bitmapDrawable = (BitmapDrawable) drawable;

```

```

Bitmap largeIcon = bitmapDrawable.getBitmap();

// Get the NotificationManager service
NotificationManager nm = (NotificationManager)
getSystemService(NOTIFICATION_SERVICE);

if (nm == null) {
    return; // If NotificationManager is null, exit
}

// Create a notification
NotificationCompat.Builder builder;

// Check if the device runs Android O or higher (API level 26+)
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
    // Create NotificationChannel for Android O and above
    NotificationChannel channel = new NotificationChannel(
        CHANNEL_ID,
        "My Channel",
        NotificationManager.IMPORTANCE_HIGH
    );
    channel.setDescription("Channel for general notifications");

    // Register the channel with the system
    nm.createNotificationChannel(channel);

    // Build the notification for Android O and above
    builder = new NotificationCompat.Builder(this, CHANNEL_ID)
        .setLargeIcon(largeIcon)
        .setSmallIcon(R.drawable.img)
        .setContentTitle("New Message")
        .setContentText("New message from jeet!")
        .setPriority(NotificationCompat.PRIORITY_HIGH)
        .setSubText("Click to see more")
        .setAutoCancel(true); // Dismiss the notification when clicked
} else {
    // Build the notification for pre-Oreo devices
    builder = new NotificationCompat.Builder(this)
        .setLargeIcon(largeIcon)
        .setSmallIcon(R.drawable.img)
        .setContentTitle("New Message")
        .setContentText("New message from jeet!")
        .setPriority(NotificationCompat.PRIORITY_HIGH)
        .setSubText("Click to see more")
        .setAutoCancel(true); // Dismiss the notification when clicked
}

```

```

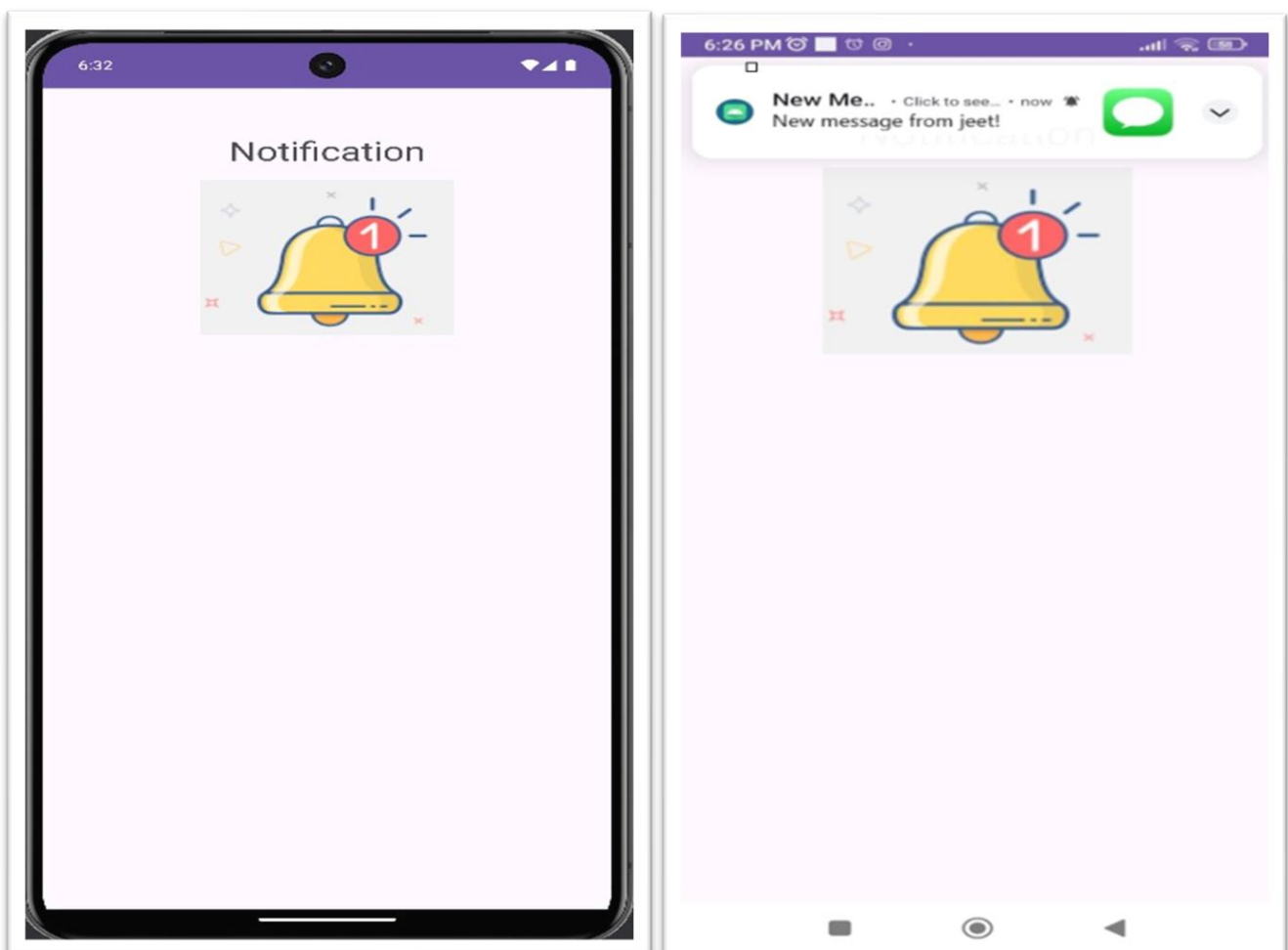
    }

    // Issue the notification
    nm.notify(NOTIFICATION_ID, builder.build());
}

// Handle the result of the permission request
@Override
public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions, grantResults);
    if (requestCode == REQUEST_NOTIFICATION_PERMISSION) {
        if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            showNotification();
        }
    }
}
}

```

Output:



Practical 14

Aim: Get and Save User Preferences

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center_horizontal"
        tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:layout_alignParentBottom="true"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="410dp"
        android:gravity="center"
        android:text="REGISTER"
        android:textSize="25dp" />

    <EditText
        android:id="@+id/enroll"
        android:layout_width="match_parent"
        android:layout_height="74dp"
        android:layout_alignParentBottom="true"
        android:layout_marginBottom="306dp"
        android:hint="Enter Enrollment"
        android:inputType="text" />

    <EditText
        android:id="@+id/registerpassword"
        android:layout_width="match_parent"
        android:layout_height="71dp"
        android:layout_alignParentBottom="true"
        android:layout_marginBottom="201dp"
        android:hint="Enter Password"
        android:inputType="textPassword" />
```

```

<Button
    android:id="@+id/registerSubmitBtn"
    android:layout_width="match_parent"
    android:layout_height="82dp"
    android:layout_alignParentBottom="true"
    android:layout_marginBottom="78dp"
    android:text="submit" />

<ImageView
    android:id="@+id/imageView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="108dp"
    android:layout_marginBottom="470dp"
    app:srcCompat="@drawable/regi" />
</RelativeLayout>

```

MainActivity.java:

```

package com.example.p13;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Check if user is already registered
        UserPreference userPreferences = new UserPreference(MainActivity.this);
        if (userPreferences.isUserRegistered()) {
            // If the user is already registered, go directly to login
            Intent intent = new Intent(MainActivity.this, Activity_login.class);
            startActivity(intent);
            finish(); // Close registration activity
            return;
        }

        setContentView(R.layout.activity_main);
    }
}

```

```

EditText name, pass;
name = findViewById(R.id.enroll);
pass = findViewById(R.id.registerpassword);
Button submit = findViewById(R.id.registerSubmitBtn);

submit.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        if (name.getText().toString().isEmpty() || pass.getText().toString().isEmpty()) {
            Toast.makeText(MainActivity.this, "Please enter all details",
                Toast.LENGTH_SHORT).show();
        } else {
            userPreferences.registerUser(name.getText().toString(),
                pass.getText().toString());
            Toast.makeText(MainActivity.this, "Registration Successful! Please log in.",
                Toast.LENGTH_SHORT).show();
            Intent intent = new Intent(MainActivity.this, Activity_login.class);
            startActivity(intent);
            finish(); // Close registration activity
        }
    }
});
}
}

```

Activity_login.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="53dp"
        android:layout_alignParentBottom="true"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="374dp"
        android:gravity="center"
        android:text="LOGIN"
    >

```

```

        android:textSize="25dp" />

<EditText
    android:id="@+id/loginenroll"
    android:layout_width="match_parent"
    android:layout_height="67dp"
    android:layout_alignParentBottom="true"
    android:layout_marginBottom="246dp"
    android:hint="Enter Enrollment"
    android:inputType="text" />

<EditText
    android:id="@+id/loginpassword"
    android:layout_width="match_parent"
    android:layout_height="65dp"
    android:layout_alignParentBottom="true"
    android:layout_marginBottom="144dp"
    android:hint="Enter Password"
    android:inputType="textPassword" />

<Button
    android:id="@+id/loginSubmitBtn"
    android:layout_width="match_parent"
    android:layout_height="63dp"
    android:layout_alignParentBottom="true"
    android:layout_marginBottom="45dp"
    android:text="login" />

<ImageView
    android:id="@+id/imageView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="90dp"
    android:layout_marginBottom="466dp"
    app:srcCompat="@drawable/regi" />

</RelativeLayout>

```

Activity_login.java:

```

package com.example.p13;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;

```

```

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class Activity_login extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        EditText name, pass;
        name = findViewById(R.id.loginenroll);
        pass = findViewById(R.id.loginpassword);
        Button submit = findViewById(R.id.loginSubmitBtn);

        submit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                UserPreference userPreferences = new UserPreference(Activity_login.this);
                boolean loginSuccessful = userPreferences.verifyUser(name.getText().toString(),
pass.getText().toString());

                if (loginSuccessful) {
                    // If login is successful, navigate to SecondActivity
                    Intent intent = new Intent(Activity_login.this, SecondActivity.class);
                    intent.putExtra("username", name.getText().toString()); // Pass the username
                    startActivity(intent);
                    finish(); // Optional: Close login activity to prevent going back
                }
            }
        });
    }
}

```

Activity_second.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"

```

```

android:padding="16dp">

<TextView
    android:id="@+id/welcomeTextView"
    android:layout_width="309dp"
    android:layout_height="139dp"

    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginStart="44dp"
    android:layout_marginTop="417dp"
    android:layout_marginEnd="42dp"
    android:layout_marginBottom="159dp"
    android:text="Welcome to app"
    android:textSize="35dp"
    android:textStyle="bold" />

<ImageView
    android:id="@+id/imageView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="82dp"
    android:layout_marginBottom="546dp"
    app:srcCompat="@drawable/regi" />

</RelativeLayout>

```

SecondActivity.java:

```

package com.example.p13;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}

```

```

// Retrieve the username from the Intent
String username = getIntent().getStringExtra("username");

TextView welcomeText = findViewById(R.id.welcomeTextView);

// Set the welcome message with the username
welcomeText.setText("Welcome to the app, " + username + "!");
}

}

```

UserPreference.java:

```

package com.example.p13;

import android.content.Context;
import android.content.SharedPreferences;
import android.widget.Toast;

public class UserPreference {
    private SharedPreferences userSession;
    private SharedPreferences.Editor editor;
    private Context context;
    private static final String KEY_NAME = "enroll";
    private static final String KEY_PASS = "userpass";
    private static final String IS_REGISTERED = "isRegistered";

    public UserPreference(Context context) {
        this.context = context;
        userSession = context.getSharedPreferences("userLoginSession",
Context.MODE_PRIVATE);
        editor = userSession.edit();
    }

    // Register user and set registration flag
    public void registerUser(String name, String pass) {
        editor.putString(KEY_NAME, name);
        editor.putString(KEY_PASS, pass);
        editor.putBoolean(IS_REGISTERED, true);
        editor.commit();
    }

    // Verify user credentials and return true if login is successful
    public boolean verifyUser(String name, String pass) {
        String storedName = userSession.getString(KEY_NAME, null);
        String storedPass = userSession.getString(KEY_PASS, null);
    }
}

```

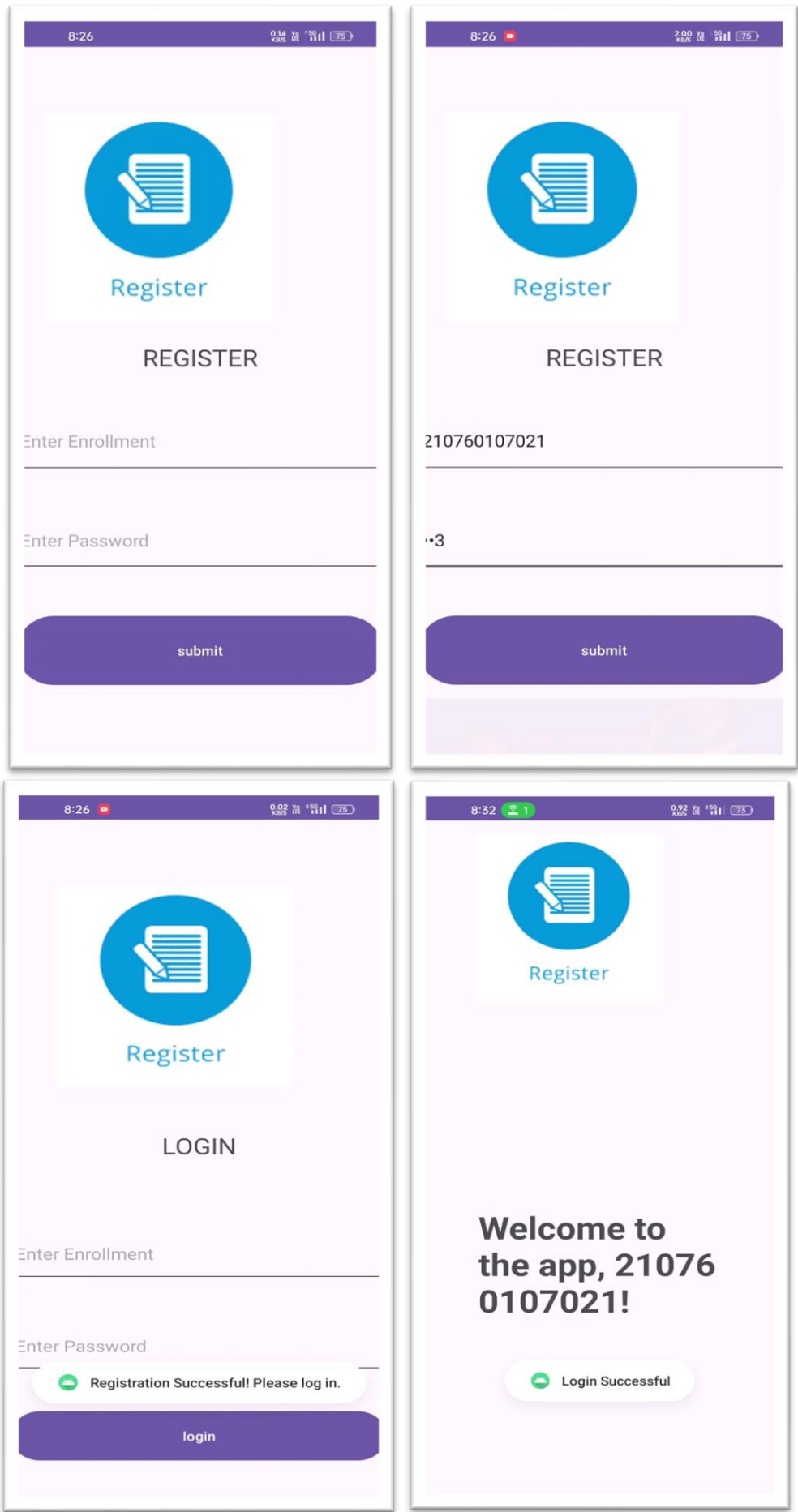
```

    if (storedName != null && storedPass != null) {
        if (name.equals(storedName) && pass.equals(storedPass)) {
            Toast.makeText(context, "Login Successful", Toast.LENGTH_SHORT).show();
            return true; // Login successful
        } else {
            Toast.makeText(context, "Invalid Enrollment or password",
Toast.LENGTH_SHORT).show();
            return false; // Login failed
        }
    } else {
        Toast.makeText(context, "Invalid Enrollment or password",
Toast.LENGTH_SHORT).show();
        return false; // Login failed
    }
}

// Check if user is already registered
public boolean isUserRegistered() {
    return userSession.getBoolean(IS_REGISTERED, false);
}
}

```

Output:



Practical 15

Aim: Make a use of android system

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    android:gravity="center_horizontal" >

    <Button
        android:id="@+id/camera_button"
        android:layout_width="100dp"
        android:layout_height="50dp"
        android:layout_marginTop="20dp"
        android:text="Camera" />

    <ImageView
        android:id="@+id/click_image"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:layout_margin="20dp"
    />

</LinearLayout>
```

MainActivity.java:

```
package com.ssasit.pr15;

import android.content.Intent;
import android.content.pm.PackageManager;
import android.graphics.Bitmap;
import android.os.Bundle;
import android.provider.MediaStore;
import android.view.View;
import android.widget.*;
import android.Manifest;
```

```

import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {
    Button btnCapture;
    ImageView imgCapture;
    int CAM_CODE=100;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnCapture=findViewById(R.id.camera_button);
        imgCapture=findViewById(R.id.click_image);

        btnCapture.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (!checkcamPermission()){
                    askPermission();
                }else {
                    captureImage();
                }
            }
        });
    }
    private void captureImage() {
        Intent camIntent=new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
        startActivityForResult(camIntent,CAM_CODE);
    }

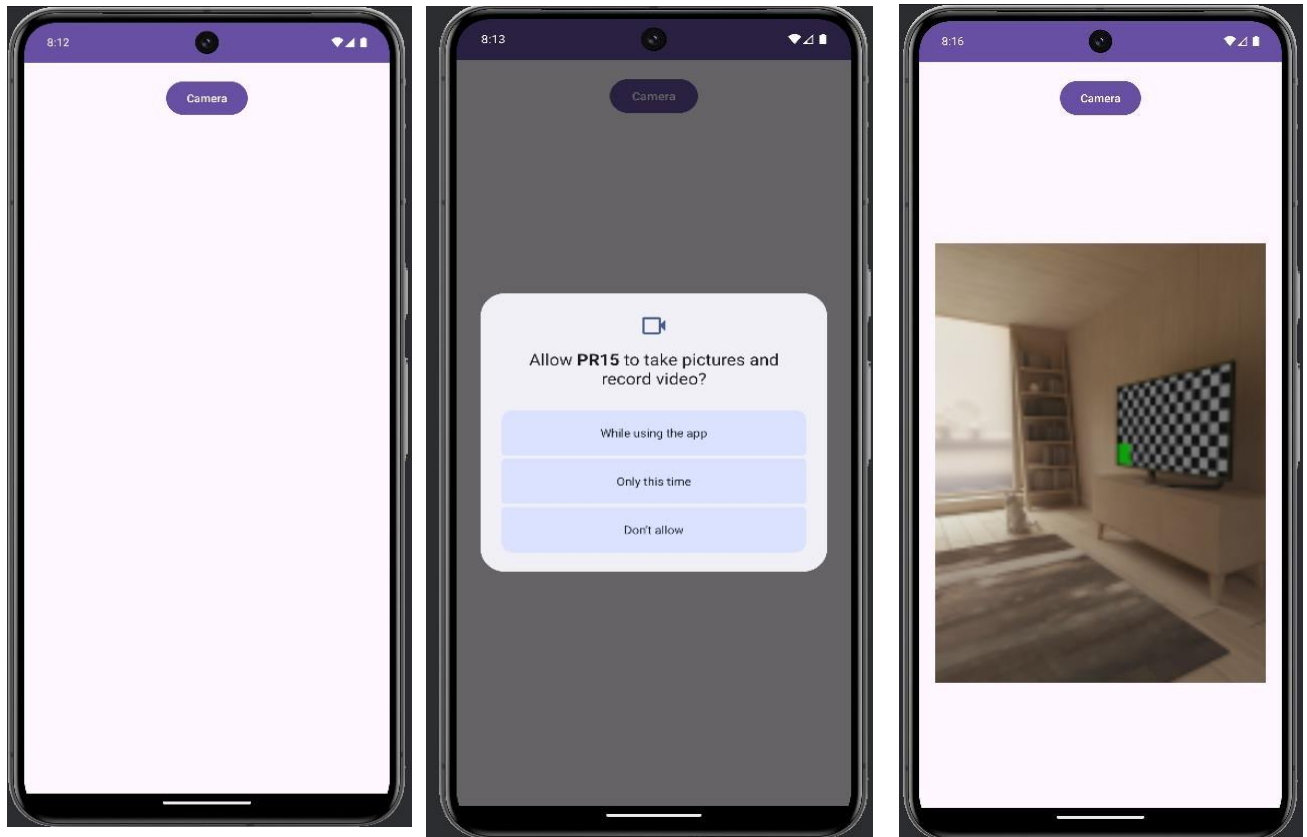
    private boolean checkcamPermission() {
        return ContextCompat.checkSelfPermission(this, Manifest.permission.CAMERA)
            == PackageManager.PERMISSION_GRANTED;
    }

    private void askPermission() {
        ActivityCompat.requestPermissions(this, new
            String[]{Manifest.permission.CAMERA},CAM_CODE);
    }
    @Override
    protected void onActivityResult(int requestCode, int resultCode, @Nullable Intent data) {
        if (requestCode==CAM_CODE &&resultCode==RESULT_OK){
            Bitmap image=(Bitmap) data.getExtras().get("data");

```

```
        imgCapture.setImageBitmap(image);  
    }  
    super.onActivityResult(requestCode, resultCode, data);  
}  
}
```

Output:



Practical 16

Aim: Using location service get the current location and display in TextView

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    android:padding="20dp">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Shree Swami Atmanand Saraswati Institute of Technology"
        android:gravity="center"
        android:fontFamily="sans-serif-black"
        android:textSize="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:fontFamily="sans-serif-black"
        android:text="Latitude:" />
    <TextView
        android:id="@+id/latTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Latitude will be here! "
        android:textColor="#f5f5f5" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:fontFamily="sans-serif-black"
        android:text="Longitude:" />
    <TextView
        android:id="@+id/lonTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Longitude will be here! "
```

```

        android:textColor="#f5f5f5" />

<Button
    android:id="@+id/btnloc"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Fetch Current Location"/>

</LinearLayout>

```

MainActivity.java:

```

package com.ssasit.pr16;

import android.annotation.SuppressLint;
import android.content.Context;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.os.Looper;
import android.provider.Settings;
import android.view.View;
import android.widget.*;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationCallback;
import com.google.android.gms.location.LocationRequest;
import com.google.android.gms.location.LocationResult;
import com.google.android.gms.location.LocationServices;
import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import android.Manifest;

public class MainActivity extends AppCompatActivity {
    FusedLocationProviderClient mFusedLocationClient;
    TextView latitudeTextView, longitTextView;
    Button loc_fetch;
    int PERMISSION_ID = 44;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

```

```

setContentView(R.layout.activity_main);

latitudeTextView = findViewById(R.id.latTextView);
longitTextView = findViewById(R.id.lonTextView);
loc_fetch=findViewById(R.id.btnloc);

mFusedLocationClient = LocationServices.getFusedLocationProviderClient(this);
loc_fetch.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        getLastLocation();
    }
});

}

@SuppressLint("MissingPermission")
private void getLastLocation() {
    if (checkPermissions()) {
        if (isLocationEnabled()) {

            mFusedLocationClient.getLastLocation().addOnCompleteListener(new
OnCompleteListener<Location>() {
                @Override
                public void onComplete(@NonNull Task<Location> task) {
                    Location location = task.getResult();
                    if (location == null) {
                        requestNewLocationData();
                    } else {
                        latitudeTextView.setText(location.getLatitude() + "");
                        longitTextView.setText(location.getLongitude() + "");
                    }
                }
            });

        } else {
            Toast.makeText(this, "Please turn on" + " your location...",
Toast.LENGTH_LONG).show();
            Intent intent = new
                Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS);
            startActivity(intent);
        }
    } else {
        requestPermissions();
    }
}

@SuppressLint("MissingPermission")
private void requestNewLocationData() {
    LocationRequest mLocationRequest = new LocationRequest();
    mLocationRequest.setPriority(LocationRequest.PRIORITY_HIGH_ACCURACY);

```

```

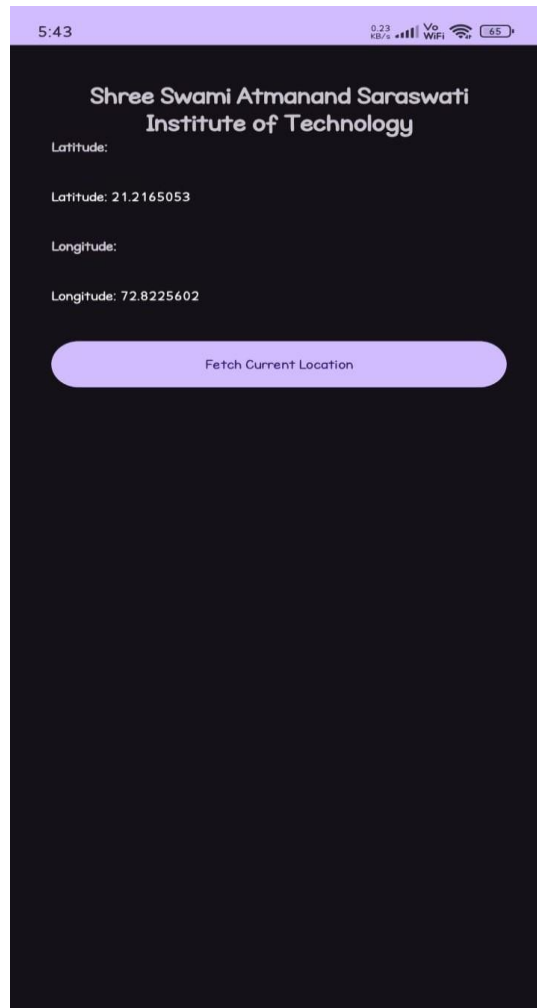
        mLocationRequest.setInterval(5);
        mLocationRequest.setFastestInterval(0);
        mLocationRequest.setNumUpdates(1);
        mFusedLocationClient = LocationServices.getFusedLocationProviderClient(this);
        mFusedLocationClient.requestLocationUpdates(mLocationRequest,
mLocationCallback, Looper.myLooper());
    }
    private LocationCallback mLocationCallback = new LocationCallback() {
        @Override
        public void onLocationResult(LocationResult locationResult) {
            Location mLastLocation = locationResult.getLastLocation();
            latitudeTextView.setText("Latitude: " + mLastLocation.getLatitude() + "");
            longitTextView.setText("Longitude: " + mLastLocation.getLongitude() + "");
        }
    };
    private boolean checkPermissions() {
        return
ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_COARSE_LOCATI
ON) ==

PackageManager.PERMISSION_GRANTED&&ActivityCompat.checkSelfPermission(this,
        Manifest.permission.ACCESS_FINE_LOCATION) ==
        PackageManager.PERMISSION_GRANTED;
    }
    private void requestPermissions() {
        ActivityCompat.requestPermissions(this, new String[]{
            Manifest.permission.ACCESS_COARSE_LOCATION,
            Manifest.permission.ACCESS_FINE_LOCATION}, PERMISSION_ID);
    }
    private boolean isLocationEnabled() {
        LocationManager locationManager = (LocationManager)
            getSystemService(Context.LOCATION_SERVICE);
        return locationManager.isProviderEnabled(LocationManager.GPS_PROVIDER) ||
            locationManager.isProviderEnabled(LocationManager.NETWORK_PROVIDER);
    }
    @Override
    public void
onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == PERMISSION_ID) {
            if (grantResults.length> 0 &&grantResults[0] ==
                PackageManager.PERMISSION_GRANTED) {
                getLastLocation();
            }
        }
    }
    @Override
    public void onResume() {
        super.onResume();

```

```
    if (checkPermissions()) {  
        getLastLocation();  
    }  
}  
}
```

Output:



Practical 17

Aim: Display the use of animations

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    android:gravity="center_horizontal">
    <ImageView
        android:id="@+id/imageview"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_marginTop="20dp"
        android:contentDescription="@string/app_name"
        android:src="@drawable/ic_launcher_background"
    />
    <Button
        android:id="@+id/animatedButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blink"
        android:layout_marginTop="20dp"
    />
    <Button
        android:id="@+id/rotateButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Rotate"
        android:layout_marginTop="20dp"
    />
</LinearLayout>
```

MainActivity.java:

```

package com.ssasit.pr17;

import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button blinkBTN = findViewById(R.id.animatedButton);
        Button rtBTN = findViewById(R.id.rotateButton);
        ImageView imageView = findViewById(R.id.imageview);

        blinkBTN.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Animation animation = AnimationUtils.loadAnimation(getApplicationContext(),
                    R.anim.blink);
                imageView.startAnimation(animation);
            }
        });
        rtBTN.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Animation animation = AnimationUtils.loadAnimation(getApplicationContext(),
                    R.anim.rotate);
                imageView.startAnimation(animation);
            }
        });
    }
}

```

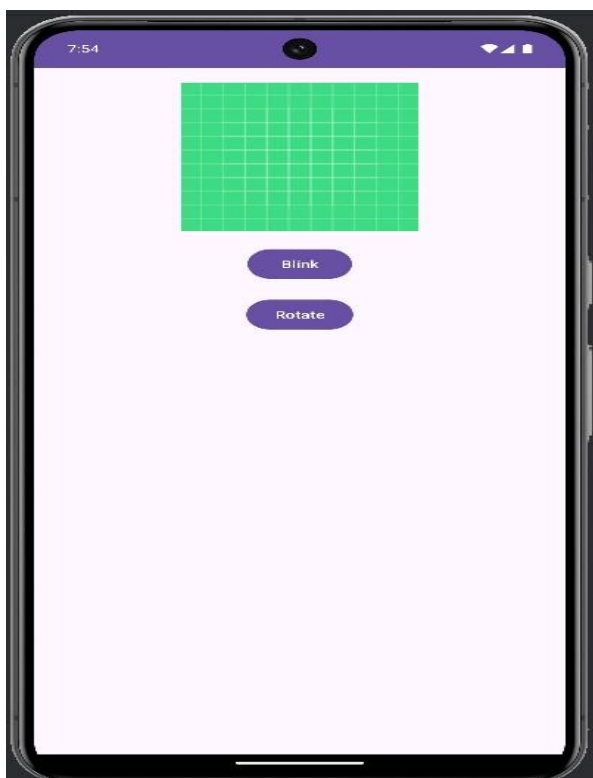
Animation Files:

blink.xml

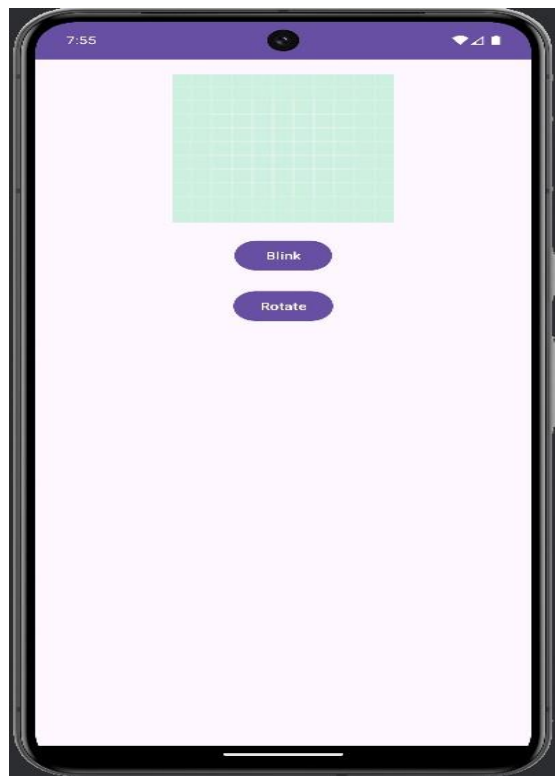
```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <alpha android:fromAlpha="0.0"
    android:toAlpha="1.0"
    android:interpolator="@android:anim/accelerate_interpolator"
    android:duration="600"
    android:repeatMode="reverse"
    android:repeatCount="infinite"/>
</set>
```

rotate.xml

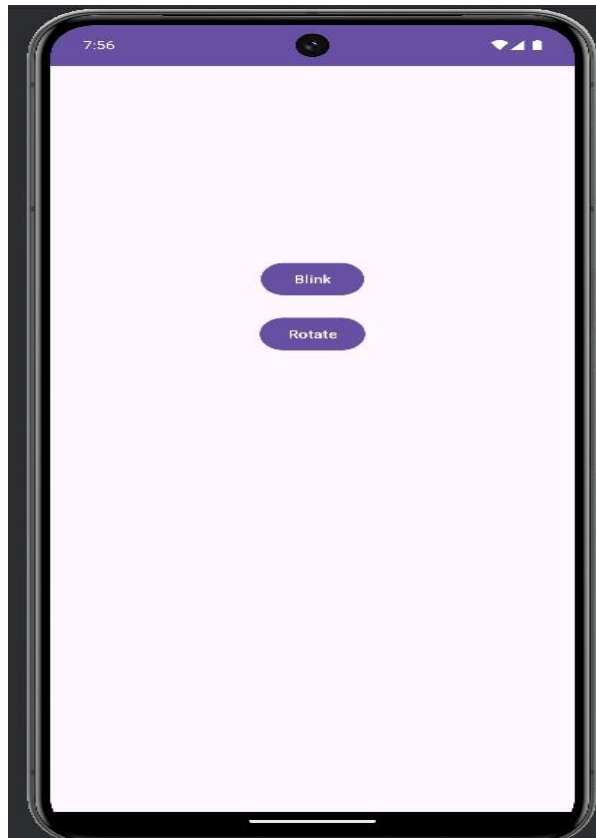
```
<?xml version="1.0" encoding="UTF-8"?>
<rotate xmlns:android="http://schemas.android.com/apk/res/android"
  android:duration="1600"
  android:fromDegrees="0"
  android:interpolator="@android:anim/linear_interpolator"
  android:pivotX="50%"
  android:pivotY="50%"
  android:repeatCount="infinite"
  android:toDegrees="358" />
```

Output:

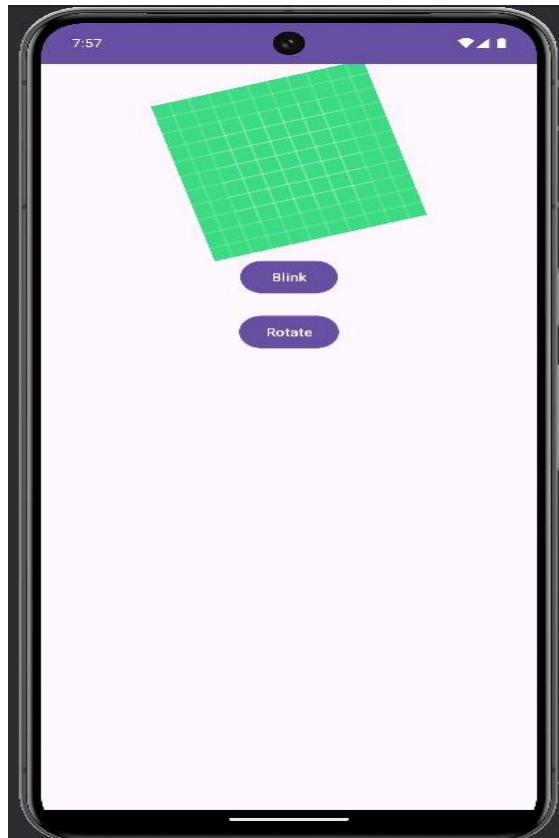
(Initial)



(Blink)



(Blink)



(Rotate)